ASSIGNMENT#1

Submitted By:

Kaleem Faryad(51)

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Submitted To: Mr.Mehtab Mushtaq

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DEPARTMENT OF CS&IT

FACULTY OF COMPUTING AND ENGINEERING

UNIVERSITY OF KOTLI AZAD JAMMU AND KASHMIR

Table of Contents

1. EVOLUTION OF OPERATING SYSTEMS 3
1.1 EARLIEST DAYS 3
1.2 KEY MILESTONES 3
1.3 TECHNOLOGICAL ADVANCEMENTS: 3
1.4 MAJOR CONTRIBUTORS 3
1.5 ANALYSIS OF TRENDS 3
2. REAL-TIME OPERATING SYSTEMS (RTOS) 3
2.1 CHARACTERISTICS 3
2.2 DIFFERENCES FROM GENERAL-PURPOSE OS 3
2.3 APPLICATIONS 4
3. MOBILE OPERATING SYSTEMS 4
3.1 DESIGN CONSIDERATIONS 4
3.2 SPECIFIC FEATURES 4
3.3 SECURITY CONCERNS 4
3.4 EVOLUTION 4
4. PRACTICAL APPLICATION ANALYSIS 4

1. EVOLUTION OF OPERATING SYSTEMS

The historical development of operating systems has transitioned from manual hardware manipulation to sophisticated automated environments.

1.1 EARLIEST DAYS

In the beginning, there were no formal operating systems; users interacted directly with hardware using plugboards.

1.2 KEY MILESTONES

The introduction of Batch Processing reduced setup time, followed by Multiprogramming and Time-Sharing in the 1960s and 70s, which optimized CPU usage.

1.3 TECHNOLOGICAL ADVANCEMENTS:

Major shifts occurred with the rise of Personal Computing and Graphical User Interfaces (GUI).

1.4 MAJOR CONTRIBUTORS

Key figures such as Ken Thompson, Dennis Ritchie (Unix), and Linus Torvalds (Linux) shaped the foundation of modern systems.

1.5 ANALYSIS OF TRENDS

Modern trends focus on cloud integration, virtualization, and mobile-first architectures.

2. REAL-TIME OPERATING SYSTEMS (RTOS)

RTOS are specialized systems designed for environments where the timing of the response is critical.

2.1 CHARACTERISTICS

The primary requirement is determinism, meaning the OS guarantees a response within a specific time constraint.

2.2 DIFFERENCES FROM GENERAL-PURPOSE OS

Unlike standard systems (like Windows) that focus on user throughput, an RTOS prioritizes task deadlines and low latency.

2.3 APPLICATIONS

These are essential in embedded systems, robotics, and time-sensitive environments such as medical devices or automotive controllers.

3. MOBILE OPERATING SYSTEMS

Operating systems for mobile devices face unique design challenges due to hardware limitations.

3.1 DESIGN CONSIDERATIONS

Developers must account for limited battery life, screen size constraints, and power management.

3.2 SPECIFIC FEATURES

Mobile platforms include features like GPS integration, touch interfaces, and specialized sensors (accelerometers).

3.3 SECURITY CONCERNS

Key concerns include app sandboxing, permission management, and data encryption to protect user privacy.

3.4 EVOLUTION

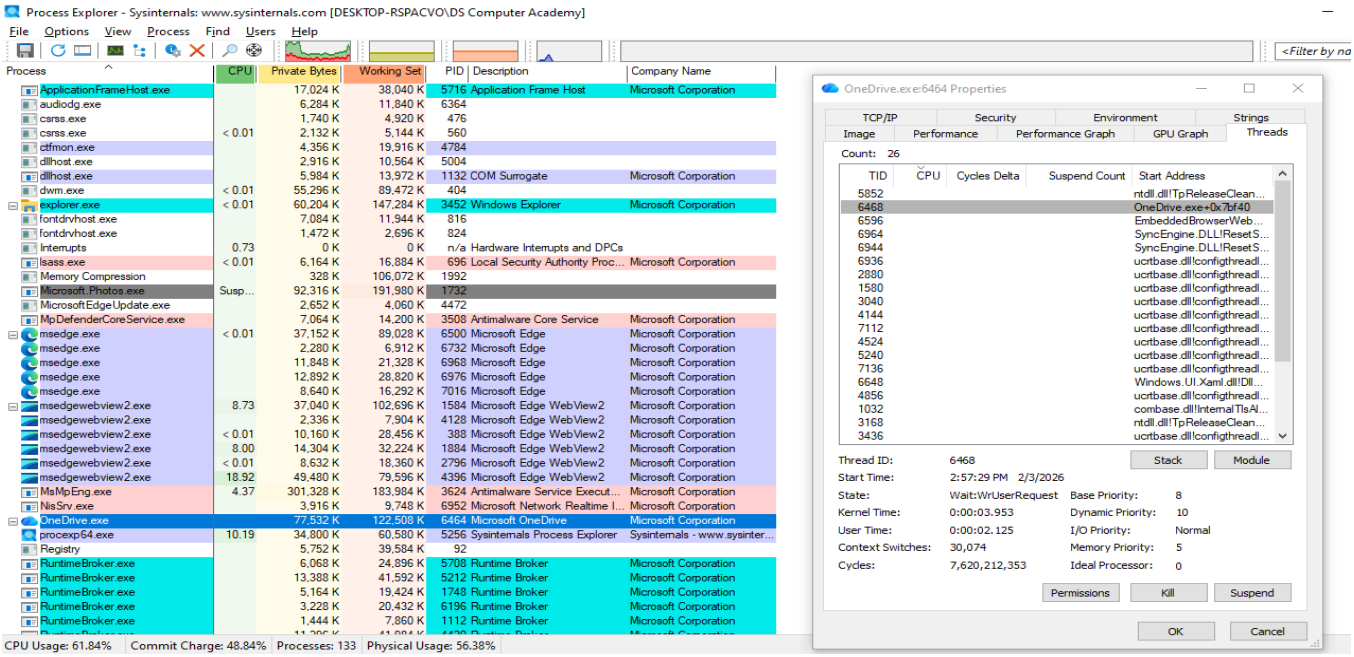
Mobile platforms have evolved from simple cellular interfaces to complex ecosystems supporting millions of apps.

4. RUN THREE DIFFERENT APPLICATION

The following data was gathered by running three applications on a local computer:

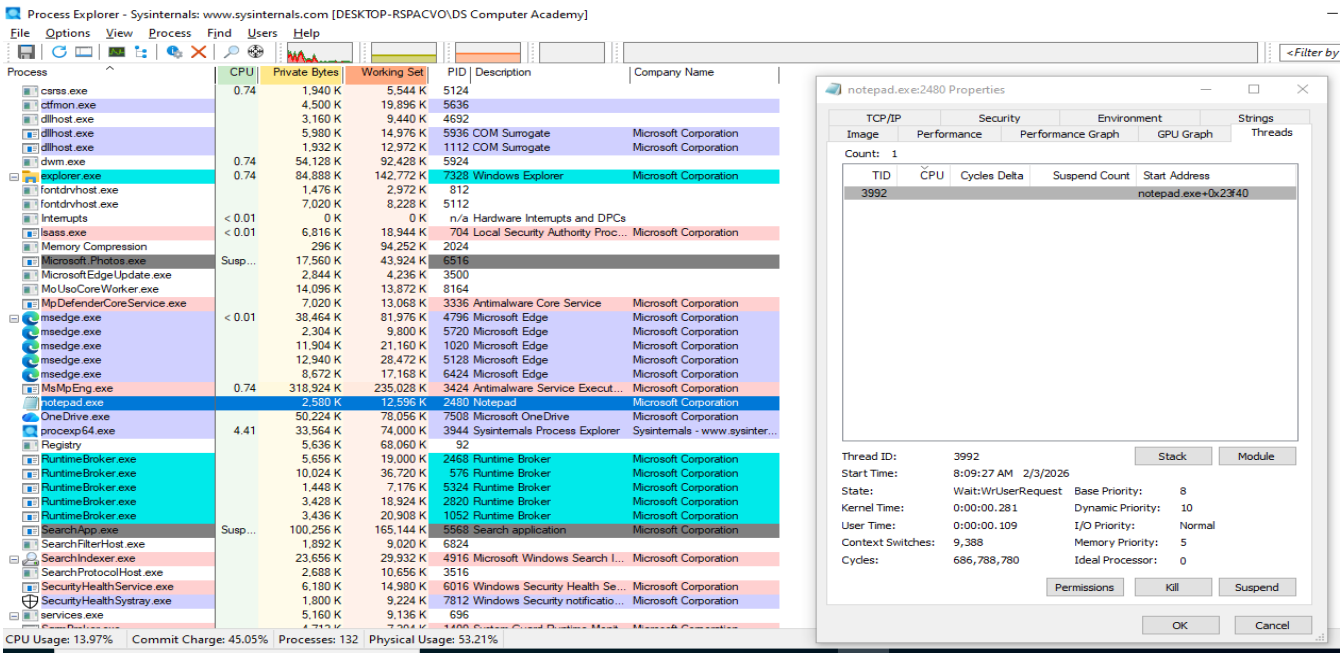
Application No:1

Application Name	**Thread**	**Thread ID**	**CPU Usage**
OverDrive.exe	28	6468	27.91%



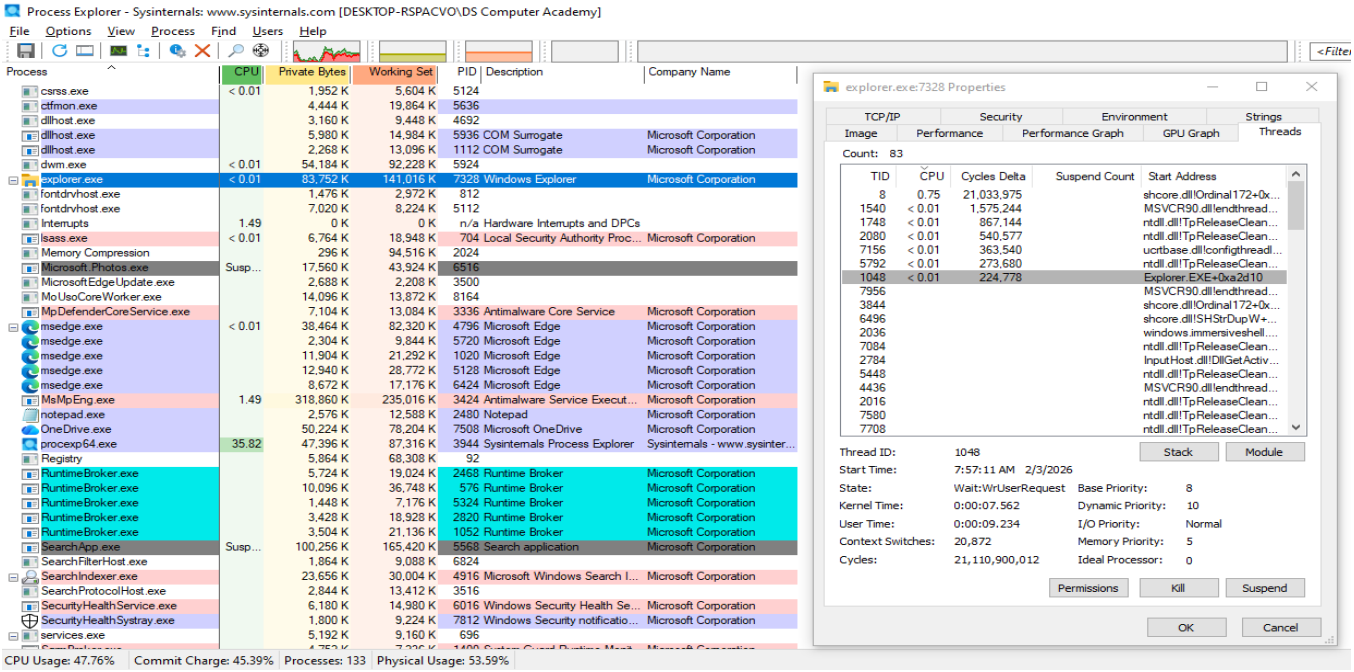
Application No:2

Application Name	Thread	Thread ID	CPU Usage
NotePade.exe	1	3992	13.97%



Application No:3

Application Name	Thread	Thread ID	CPU Usage
Explorer.exe	83	1048	47.76%



Final Table

Application Name	Thread	Thread ID	CPU Usage
Explorer.exe	83	1048	47.76%
NotePade.exe	1	3992	13.97%
Explorer.exe	83	1048	47.76%